

Re-Implementing and Comparing Clustering Algorithms for Single-Cell RNA Sequencing Data

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ABSTRACT

This article focuses on the re-implementation of the method proposed in "Quantifying the Clusterness and Trajectory-ness of Single-Cell RNA-seq Data" to evaluate the clusterness and trajectory-ness of scRNA-seq data. The goal is to provide a quantitative framework for analyzing whether scRNA-seq data exhibits distinct clustering or continuous trajectories. The method utilizes geometric metrics, including pairwise distance distributions, persistent topology analysis, and connectivity indices, to assess the underlying structure of the data. Various parameter settings and data subsets are explored to gain critical insights into the strengths and limitations of this approach. The findings of this study offer a reproducible methodology for single-cell RNA sequencing data analysis, contributing to the development of more robust computational tools for understanding cellular diversity and dynamics in biological contexts.

Keywords: scRNA-seq, clusterness, trajectory-ness, clustering, trajectory inference, persistent homology, data analysis, single-cell genomics

1 INTRODUCTION

Single-cell RNA sequencing (scRNA-seq) has revolutionized modern genomics by enabling transcriptomic profiling at the resolution of individual cells. Unlike bulk RNA sequencing, which averages gene expression over a population of thousands or millions of cells, scRNA-seq captures the variability inherent in biological systems, shedding light on cell-to-cell heterogeneity, rare subpopulations, and dynamic cellular transitions that were previously obscured. As a result, scRNA-seq has become a powerful tool for identifying new cell types, reconstructing differentiation trajectories, studying tumor microenvironments, and understanding immune responses and disease processes at an unprecedented level of granularity.

Analyzing scRNA-seq data, however, remains a computational challenge. The data is high-dimensional, sparse, and noisy, and the biological structures embedded in the expression space are often complex and nonlinear. To extract meaningful insights from such data, two dominant classes of analytical methods have emerged: clustering and trajectory inference. Clustering methods aim to group cells with similar expression profiles into distinct clusters, typically interpreted as cell types or discrete functional states. Trajectory inference methods, on the other hand, attempt to reconstruct continuous biological processes, such as cell differentiation or reprogramming, by inferring pseudotemporal progressions across cells. These methods are based on fundamentally different assumptions: clustering seeks well-separated, compact regions in expression space, while trajectory inference assumes a manifold-like structure with smooth transitions.

Despite their shared goal of revealing underlying biological structure, clustering and trajectory inference often yield divergent interpretations of the same dataset. A given dataset may appear as a set of discrete populations when visualized with clustering algorithms like Leiden or Louvain, yet exhibit smooth, continuous variation under trajectory inference tools like Monocle, Slingshot, or Diffusion Pseudotime. This interpretational inconsistency poses a critical problem: how can we objectively decide whether to treat a dataset as composed of clusters or as following a trajectory? Relying solely on subjective criteria such as visual inspection of low-dimensional embeddings or prior biological expectations can lead to biased analyses and potentially misleading conclusions.

To address this methodological gap, Lim and Qiu (2024) [1] proposed a framework for quantifying the geometric structure of scRNA-seq data. Their method evaluates the extent to which a dataset exhibits "clusterness" or "trajectory-ness" using five independent, interpretable metrics derived from geometry, topology, and graph theory.

The resulting five-dimensional score vector is projected into a two-dimensional "geometry space" using UMAP. This space is trained on a reference collection of thousands of synthetic datasets, each simulating either clean clusters, smooth trajectories, noisy hybrids, or disconnected components, allowing any new dataset to be positioned and interpreted within this learned continuum.

In this project, we aim to reproduce, evaluate, and extend the method introduced by Lim and Qiu. First, we re-implement the full pipeline using the authors' published code, verifying the reproducibility of their results on the reference datasets. Next, we explore the robustness and interpretability of the five scores by altering various experimental conditions: the level of noise in synthetic datasets, the choice of distance metrics (e.g., diffusion-based vs. Euclidean), dimensionality reduction techniques (PCA vs. diffusion maps), and clustering granularity. We further apply the framework to several real-world scRNA-seq datasets, including well-characterized developmental trajectories and immune cell atlases, in order to test whether the geometric scores provide meaningful guidance when clustering and trajectory inference disagree.

Our study serves as both a technical validation and a critical assessment of a recent methodology in single-cell data analysis. By systematically analyzing the behavior of each score under different conditions, we seek to determine their reliability, sensitivity, and practical usefulness in real analytical workflows. Ultimately, we aim to clarify under which circumstances each score is informative, how their combination can support decision-making, and whether this framework can help move scRNA-seq analysis toward more reproducible, objective, and geometry-aware practices.

2 MATERIALS AND METHODS

The implementation is publicly available on the : GitHub repository. We used the provided Jupyter notebooks as a base for the re-implementation, making necessary modifications to adapt it to our experimental setup.

In this study, we use both simulated and real single-cell RNA sequencing (scRNA-seq) datasets to evaluate the proposed metrics for quantifying clusterness and trajectoriness. The simulated datasets include four types: (1) clear clusters data, (2) clear trajectory data, (3) noisy clusters data, and (4) noisy trajectory data, each designed to test different aspects of the proposed scoring metrics. We generated these datasets using a combination of Gaussian mixture models for clusters and sine functions with added Gaussian noise for trajectories.

For real scRNA-seq datasets, we utilize publicly available datasets from various sources, including Planaria dataset, Mouse Bone-Marrow dataset, Monkey Epiblast dataset, and others. All datasets were pre-processed using the methods described in the original benchmark papers, including gene filtering, normalization, and PCA transformation to reduce dimensionality. This study proposes a method to quantify the **clusterness** and **trajectoriness** of scRNA-seq datasets based on five scoring metrics. These metrics assess the underlying geometric properties of the data and help distinguish between cluster-like and trajectory-like structures.

Scoring Metrics

Five key metrics are computed to capture the geometric structure of the data:

(1) **Distribution of Pairwise Distances:**

The entropy of the pairwise distances between all cells in the dataset is computed using **geodesic distances** derived from **Diffusion Pseudotime (DPT)** analysis. The entropy of the normalized histogram of these distances serves as a metric, where lower entropy indicates cluster-like data, and higher entropy suggests trajectory-like data.

(2) **Persistent Homology:**

Persistent homology is applied to the dataset, focusing on 0-dimensional homology, which tracks the merging of connected components as the distance threshold increases. The entropy of the recorded thresholds is calculated as the scoring metric. Datasets with distinct clusters show larger gaps in component merging, while trajectory-like data demonstrates more uniform merging.

(3) **Vector Magnitude:**

The magnitude of the vector sum, generated by over-clustering the dataset and connecting the cluster centers, is computed. This provides a measure of the directionality of the data. Datasets with strong directionality, characteristic of trajectories, are expected to exhibit larger vector magnitudes.

(4) **Ripley's K Function:**

Ripley's K function is used to analyze the spatial distribution of points in the dataset. By comparing the observed spatial distribution to a random uniform distribution, this function helps assess the degree of clustering. Cluster-like

data exhibits stronger clustering, while trajectory-like data shows weaker clustering.

(5) **Degrees of Connectivity:**

This metric quantifies the connectivity of cells based on the number of neighbors considered for each cell. In cluster-like data, cells are more likely to be connected within their respective clusters, while trajectory-like data shows broader connectivity across the dataset.

Data Preprocessing and Projection

To evaluate these metrics, we first compute the five scores for each of the simulated datasets, yielding a matrix of size **12000 x 5**, where 12000 is the number of simulated datasets and 5 corresponds to the number of scoring metrics. These scores are calculated through the **scoring** function implemented in the **utils.py** script. The function summarizes the spatial structure of both simulated and real datasets into a 5-dimensional vector, after applying a preprocessing phase to it, which consisted into a density downsampling function, followed by a normalization function. The "normalise" function in the preprocessing phase simply apply a MinMax scaling column by column, in order to transform each features of the dataset between 0 and 1. This step ensures that all dimensions contribute equally to distance-based calculations, such as in the density estimation stage, which is particularly important before computing pairwise distances or applying geometric metrics. Without normalization, features with larger numeric ranges could dominate the analysis, leading to biased interpretations of the dataset's spatial structure. Before the normalization, the preprocessing step begins with a density-based downsampling function, where points falling below the 5th percentile of local neighborhood density (**od = 0.05**) are discarded to remove isolated outliers. In contrast, all dense regions are preserved by setting the upper density threshold **td = 1**. These thresholds were chosen empirically to balance denoising and data preservation, while maintaining the geometric integrity of the dataset.

Violin plots were used to visualize the distribution of each geometric score across four simulated dataset types. They allow direct inspection of how well each metric separates cluster-like and trajectory-like data, demonstrating that each score captures meaningful geometric features.

The computed score matrix was then projected into a two-dimensional space using **UMAP** (Uniform Manifold Approximation and Projection). This step generates a **12000 x 2** matrix, representing the "clusterness" and "trajectoriness" of each dataset. The following UMAP parameters are used for the projection:

- **Distance Metric:** Euclidean distance.
- **Number of Neighbors:** 30.
- **Minimum Distance:** 0.6.

The resulting two-dimensional visualization allows for the assessment of the geometric structure of each dataset and provides a visual representation of the cluster-like and trajectory-like properties.

Parameter Exploration and Preprocessing

The parameters of 5 and 100 percent that were in the downsampling phase, were used by the authors in order to discard outliers, while preserving the main geometry. In order to better understand the meaning of these parameters, several other scoring function were applied, with other parameters for their preprocessing phase (See the section **results** for the comparison of those) The resulting scores for each simulation were stored in .npy files — NumPy’s binary format — for efficient saving and loading. Each .npy file corresponds to a given parameter setting, making it easier to compare the influence of preprocessing choices on the five geometric scores across the 12,000 simulated datasets. This comparison was done with the help of violin plots.

To gain a better understanding of how different preprocessing techniques affect the UMAP projection, various normalization methods and UMAP parameters are explored. This includes varying the minimum distance, distance metric, number of neighbors, and testing alternative projection functions (such as PCA). The goal is to identify how changes in these parameters influence the final geometric layout.

Geometric Classification

After obtaining the UMAP projections, **K-Nearest Neighbors (KNN)** classification is applied to predict whether each real dataset exhibits a cluster-like or trajectory-like geometry based on its position in the UMAP projection space. The classifier is trained on the simulated datasets.

To evaluate the robustness of this classification approach, an alternative classification model, **Random Forest**, is also explored. This model creates decision trees to partition the space into regions based on the five scores. Random Forest, by utilizing a greater number of trees, is able to model more complex, non-linear decision boundaries, improving classification accuracy compared to KNN.

The performance of both models is compared to assess whether the separation between cluster-like and trajectory-like geometries remains consistent when using a more complex decision boundary.

Application to Real scRNA-seq Data

Once the geometric framework is constructed and validated using the simulated datasets, the method is extended to real **scRNA-seq** datasets. A collection of **169 scRNA-seq datasets** from previous benchmarks is used, where the presumed geometries of the datasets are classified based on biological intuition and experimental design.

- **Cluster-like geometries:** These include datasets with distinct clusters, organs, or disconnected graph structures.
- **Trajectory-like geometries:** These include datasets exhibiting tree-like structures, convergence, linear paths, cycles, acyclic graphs, bifurcations, or multifurcations.

The five scoring metrics are computed for each real datasets, storing the result in `'real_data_scores.npy'` and their projections are mapped into the UMAP space created from the simulated data. This allows us to assess whether the real datasets align more closely with cluster-like or trajectory-like geometries. Before applying this mapping for real datasets to the UMAP landscape of the simulated one, Principal Component Analysis (PCA) was used in this PDF to reduce the dimensionality of real datasets to 20 components before

scoring. Here, in order to try to justify this choice of number of dimensions kept, we focused on the "filtered_TS_Bladder" dataset of the "data_scHuman" directory. This dataset comes from single-cell RNA sequencing (scRNA-seq) of human bladder cells, where each row corresponds to a single cell and each column to a gene. In this dataset, most values are zero because in a given cell, only a small fraction of genes are actively expressed. To investigate how much biological information is retained when reducing dimensionality, we applied PCA and analyzed how compression impacts (in general) the structure of the data. To do so, we measured the mean Kullback-Leibler divergence between the original data and its reconstruction after PCA with varying numbers of components. (see in section results)

Intra-dataset Cluster Analysis

In addition to evaluate each real dataset as a whole, the authors further analyzed their internal structure by decomposing each dataset into sub-clusters. For this, they applied Louvain clustering to each of the 169 scRNA-seq datasets individually, resulting to a collection of smaller subsets.

Each of these sub-clusters was then scored using the same five geometric metrics described earlier and projected into the same UMAP space constructed from the simulated datasets.

Final Validation and Visualization

Finally, the real datasets are visualized in the UMAP space as scatter-plots. This allows for visual verification of whether the real datasets exhibit cluster-like or trajectory-like geometric properties, as compared to the simulated datasets. The validity of the classification is confirmed by examining the consistency of the geometric landscape when applied to new real-world scRNA-seq data.

3 RESULTS

UMAP vs. PCA for Geometric Embedding

To assess how well the five computed metrics capture meaningful geometric differences across datasets, we first reproduced the two-dimensional embedding of the simulated dataset collection using both UMAP and PCA. This embedding serves as a reference space where both synthetic and real datasets can be projected and compared.

Figure 1 compares the structure of this geometry space under UMAP (left) and PCA (right). In both cases, each point corresponds to a simulated dataset represented by its five-dimensional vector of clusterness and trajectoriness scores. Each tile displays the 2D projection of the dataset it represents (e.g., clean clusters, trajectories, noisy variants), offering a visual intuition of the underlying structures.

The UMAP projection, computed with 30 neighbors and default parameters, reveals a rich, organized layout where datasets exhibiting similar geometric properties are grouped together. For instance, well-separated clusters occupy one region of the map, while smooth trajectories form a distinct continuous zone. Noisy or hybrid simulations tend to occupy intermediate regions, reflecting their mixed structure. Importantly, UMAP preserves both local continuity and global structure, highlighting fine distinctions between datasets that share global features but differ in local geometry. By contrast,

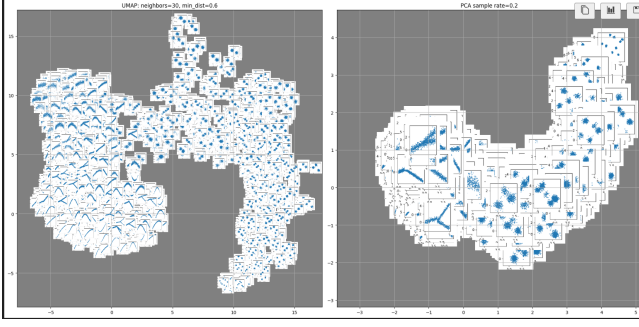


Figure 1: Comparison of the simulated geometric landscape using UMAP (left) and PCA (right). Each tile shows a 2D projection of a simulated dataset. UMAP reveals better topological separation between dataset types, while PCA fails to clearly distinguish different geometric classes.

the PCA projection (right panel) fails to separate datasets according to their structural types. Clusters and trajectories are more entangled, and local neighborhoods do not correspond clearly to specific simulation classes. This loss of organization likely stems from PCA’s linear nature, which cannot fully capture the nonlinear relationships encoded by the five geometric metrics.

These results confirm that UMAP is a more suitable tool than PCA for visualizing the geometry of scRNA-seq datasets based on clusterness and trajectory-ness scores. It better reflects the topological diversity of the simulations and enables clearer interpretations when projecting new real-world datasets into this learned space.

Classifying Trajectories with Random Forest

To further analyze the geometric interpretation space, we compared two methods for predicting the “trajectory-likeness” of a dataset within the UMAP-embedded geometry space: interpolation-based smoothing versus supervised classification using a Random Forest model.

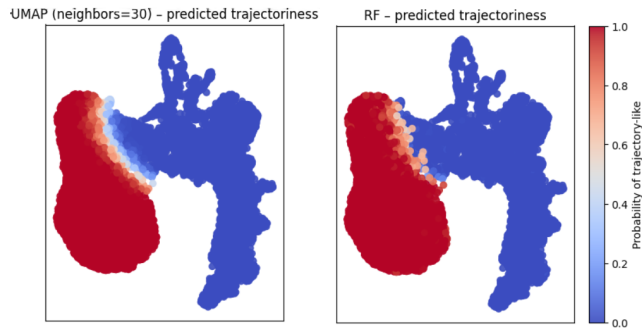


Figure 2: Predicted “trajectory-likeness” in the geometry space. Top: Kernel-smoothed trajectory probability using UMAP neighbors. Bottom: Probabilities predicted by a Random Forest classifier trained on simulation labels. The RF model yields more defined boundaries between cluster-like and trajectory-like regions.

Figure 2 (top) shows the trajectory probability over the UMAP landscape obtained using kernel smoothing on the UMAP coordinates (with 30 neighbors). This approach provides a continuous and locally smooth gradient, where datasets on the left exhibit high predicted trajectory-likeness (red), and those on the right appear more cluster-like (blue). The transition between these two regimes is gradual, especially in regions where hybrid or noisy structures are common.

Figure 2 (bottom) presents the output of a Random Forest classifier trained to distinguish cluster-like from trajectory-like simulated datasets, using the five-dimensional score vectors. The prediction probabilities are then projected back onto the same UMAP space. Compared to the smooth UMAP-based method, the Random Forest produces sharper and more defined boundaries between geometric classes. These clearer transitions may result from the model’s ability to capture complex nonlinear interactions between metrics and to impose discrete decision boundaries.

Overall, both approaches reveal a consistent global structure in the geometry space, with similar regions identified as highly trajectory-like. However, the Random Forest model appears more sensitive to subtle differences in score combinations, which may improve interpretability and class separability when projecting new datasets. This suggests that integrating machine learning models with geometric embeddings can enhance the granularity and precision of scRNA-seq structure characterization.

Projection of Scores Across Embedding Spaces

To further understand how each of the five metrics contributes to the geometry landscape, we visualized the individual score values across the two-dimensional projection space using both UMAP (top row) and PCA (bottom row) embeddings (Figure 3). Each column corresponds to one of the five scores: Pairwise Distance Entropy (P-dist), Persistent Homology, Vector Magnitude, Ripley’s K, and Degree of Separation.

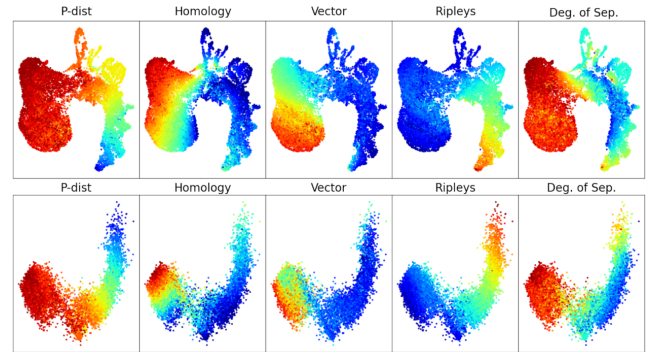


Figure 3: Spatial distribution of geometric scores on UMAP (top) and PCA (bottom) embeddings. Each column corresponds to one of the five scores: P-dist, Homology, Vector, Ripley’s K, and Degree of Separation. Score patterns remain consistent across embeddings, despite structural differences.

Despite notable differences in the overall shape of the embeddings — with PCA showing more diffuse and less topologically

coherent structures compared to UMAP — the relative spatial distribution of scores remains consistent across both embeddings. For instance:

The P-dist and Homology scores gradually increase from right to left, consistently identifying the trajectory-like region.

The Vector score shows high intensity in the lower-left region in both projections, suggesting a directionally coherent subset of datasets.

The Ripley’s K score clearly highlights cluster-like regions in the lower-right corner, especially in UMAP.

The Degree of Separation score exhibits a smooth gradient along the cluster-to-trajectory continuum in both PCA and UMAP.

This consistency across two independent embedding techniques indicates that the five scores capture intrinsic geometric features of the datasets, independent of the projection method. However, the UMAP projection offers sharper transitions and better topological organization, making it more suitable for interpretation and visualization of dataset geometry.

KL During Dimensionality Reduction with PCA

To determine a suitable trade-off between data compression and information preservation, we analyzed the information loss induced by reducing the data dimensionality using PCA. Specifically, we computed the mean Kullback-Leibler (KL) divergence between the original five-dimensional geometric score space and its reconstruction from an increasing number of principal components.

Figure 4 displays the information loss as a function of the number of PCA components. The left panel shows the KL divergence in linear scale, while the right panel uses a logarithmic scale to highlight differences at lower values. As expected, the information loss decreases monotonically with the number of principal components. However, both panels reveal that the marginal gain in information preservation diminishes rapidly after approximately 20 components.

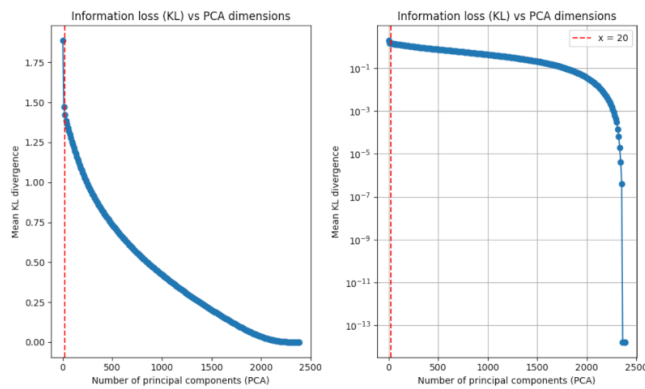


Figure 4: KL divergence between the original five-dimensional score space and PCA reconstructions with increasing dimensionality. The left panel shows linear scale; the right uses logarithmic scale to highlight the rapid drop in divergence, which plateaus beyond 20 dimensions. The red dashed line indicates a practical cutoff.

In particular, the logarithmic plot (right panel) shows a sharp drop in KL divergence between 1 and 20 dimensions, followed by a plateau where additional components contribute minimal improvement. This observation justifies the choice of a compact PCA representation (e.g., 20 components), which sufficiently captures the essential geometric structure while reducing computational cost.

The red dashed line at $x = 20$ emphasizes this practical trade-off, suggesting that a 20-dimensional embedding retains most of the geometric signal present in the full dataset with minimal information loss. This dimensionality was used in all downstream analyses involving PCA-based projections and comparisons.

Mapping Real Datasets in the Simulated Geometric Landscape

Following the identification of a 20-dimensional PCA embedding as a compact yet information-preserving representation of the geometric score space, we projected real scRNA-seq datasets into the simulated clusteriness–trajectoriness UMAP landscape.

Using the five geometric scores compressed via PCA to 20 dimensions, we mapped 169 real datasets onto the two-dimensional UMAP space learned from simulated data. As shown in Figure 5 (top), each red triangle represents one real dataset projected into this landscape. The embedding reveals how real data align with simulated patterns: datasets cluster along a trajectory from highly cluster-like to highly trajectory-like regions.

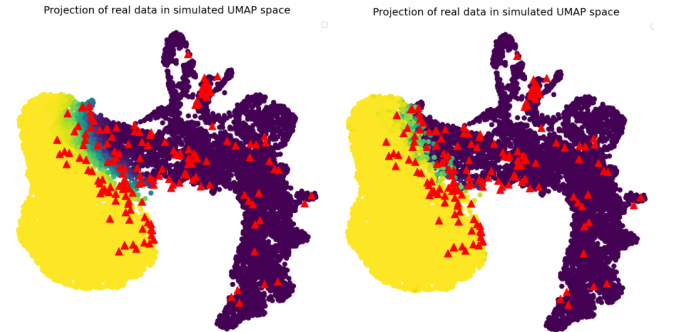


Figure 5: Projection of 169 real scRNA-seq datasets (red triangles) onto the simulated UMAP landscape using PCA-reduced (20D) geometric scores. Top: original projection. Bottom: overlay with Random Forest predictions of trajectory-like (yellow) and cluster-like (purple) regions.

To further refine this interpretation, we trained a Random Forest classifier on the simulated datasets to distinguish between cluster-like and trajectory-like geometries based on the five scores. We then applied the trained model to predict the geometry of each real dataset and visualized the output as a color-coded background in Figure 5 (bottom). Yellow regions correspond to high predicted trajectory-likeness, while dark purple indicates strong clusteriness. The projection of real datasets aligns well with this structure, with many datasets falling into transitional or ambiguous zones, reflecting diverse geometric properties.

Together, these results demonstrate the applicability of the simulated landscape as a reference space, enabling objective placement and interpretation of real-world scRNA-seq datasets based on their intrinsic geometry.

Score Distributions in Simulated vs. Real Data

To validate whether the geometric characteristics captured by the five scoring metrics are preserved when projecting real scRNA-seq datasets into the simulated UMAP landscape, we compared the distribution of each score between simulated and real datasets. Figure 6 shows violin plots of the five scores—Pairwise Distance Entropy (P-dist), Persistent Homology, Vector Magnitude, Ripley’s K, and Degree of Separation—computed for both simulated datasets and real scRNA-seq datasets. Distributions are displayed separately for two different projection strategies: K-Nearest Neighbors (KNN, top row) and Random Forest (RF, bottom row).

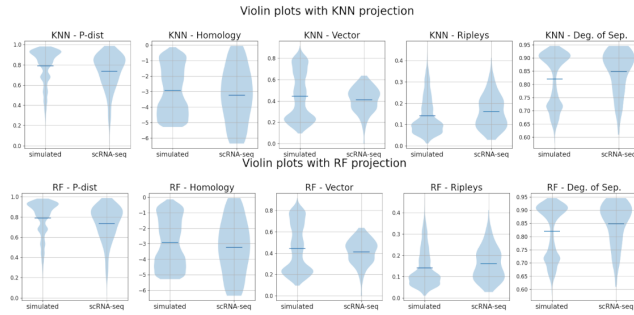


Figure 6: Distribution of geometric scores in simulated versus real scRNA-seq datasets under two projection strategies. Top: Projection via KNN-based interpolation. Bottom: Projection via Random Forest classification. In both cases, the five scores exhibit highly similar distributions between simulated and real datasets, indicating that the geometric landscape generalizes well to real-world data.

Across both projection methods, the score distributions for real datasets closely resemble those of the simulated datasets. This similarity suggests that the five-dimensional geometric representation retains its interpretability and discriminative power when applied to real-world scRNA-seq data. Furthermore, it confirms that both KNN interpolation and Random Forest classification are robust in preserving the geometric signature of the data within the UMAP space. Consequently, the simulated landscape can serve as a meaningful reference for interpreting real data.

Comparison of KNN and Random Forest Predictions

To evaluate how well the predicted geometric classifications align with biological expectations, we summarized the distribution of predicted “cluster-like” and “trajectory-like” labels across 10 types of benchmark datasets. Figure 7 shows this comparison using both KNN interpolation and Random Forest classification.

Overall, both methods produce similar trends and achieve reasonable agreement with presumed geometric labels. For instance,

	predicted clusters	predicted trajectory
acyclic_graph	1	0
bifurcation	7	6
cluster	19	13
convergence	1	0
cycle	1	1
disconnected_graph	22	3
linear	18	21
multifurcation	5	4
organ	18	10
tree	16	3

RF table:		
	predicted clusters	predicted trajectory
acyclic_graph	1	0
bifurcation	7	6
cluster	20	12
convergence	1	0
cycle	1	1
disconnected_graph	22	3
linear	17	22
multifurcation	5	4
organ	18	10
tree	17	2

Figure 7: Comparison between expected (benchmark-assigned) and predicted geometry classes for 10 types of scRNA-seq datasets. Top: Predictions via KNN smoothing on the UMAP geometry space. Bottom: Predictions via Random Forest classifier.

datasets labeled as `disconnected_graph` or `cluster` are mostly predicted as “cluster-like”, while datasets labeled as `linear` are predominantly classified as “trajectory-like”.

However, upon closer inspection, the Random Forest classifier exhibits a bias toward predicting datasets as “cluster-like”. This is particularly evident in ambiguous categories such as `tree`, `bifurcation`, and `organ`, where RF assigns more datasets to the cluster category compared to the KNN-based method. This behavior may reflect the sharper decision boundaries imposed by supervised learning, which contrast with the more continuous transitions captured by local smoothing in UMAP space.

These differences highlight the trade-off between smooth interpolation and discrete classification in modeling dataset geometry. While both methods agree on most clear-cut cases, their divergence in intermediate regions underlines the complexity of the cluster–trajectory continuum.

4 DISCUSSION

Indeed, this observation suggested that global dataset geometry can sometimes mask underlying heterogeneity or internal transitions.

This work aimed to reproduce and evaluate the geometric framework proposed by Lim and Qiu (2024), designed to quantify the “clusterness” and “trajectoriness” of single-cell RNA sequencing (scRNA-seq) datasets. Our reimplementation confirmed the relevance of the original pipeline while also highlighting several critical

aspects regarding its robustness, limitations, and practical applicability in real biological analyses.

First, our results validate that the combination of five geometric metrics — pairwise distance entropy, persistent homology, vector magnitude, Ripley’s K function, and degree of connectivity — can effectively distinguish between cluster-like and trajectory-like structures. The projection of these metrics into a UMAP-defined space, trained on simulated datasets, provides an intuitive visualization of the geometric landscape. This landscape captures subtle gradients along the cluster-ness-to-trajectory-ness continuum, moving beyond simple binary classification and demonstrating the method’s ability to reflect biological structural diversity. However, our study also revealed important methodological limitations. The robustness of the framework is highly dependent on data preprocessing. The use of PCA for dimensionality reduction to 20 components, although justified by an analysis of Kullback-Leibler (KL) divergence, may obscure fine-grained structures. Likewise, density-based downsampling and MinMax normalization, while useful for noise reduction, may introduce bias in topological analyses.

Another critical point concerns the tension between global and local interpretations of dataset geometry. As highlighted by the sub-cluster analysis (e.g., Figure 8), a dataset that appears globally trajectory-like may contain locally clustered substructures, and vice versa. This raises an important question: at what level should the geometric framework be applied — the entire dataset or individual subpopulations? This ambiguity has tangible implications for downstream analyses, such as choosing between clustering algorithms or pseudotime inference. In some cases, a global projection might obscure the underlying cellular complexity, potentially leading to misleading biological interpretations.

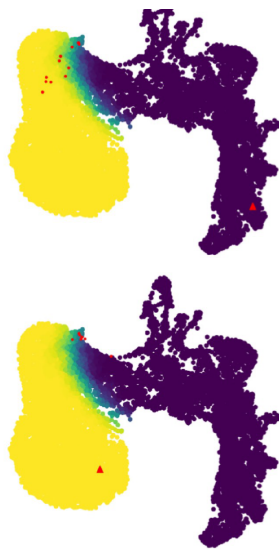


Figure 8: Projections of cluster and trajectory-like datasets, and their individual clusters, shown by red triangle and red dots respectively

Our comparison between PCA and UMAP projections of the geometric scores further demonstrates the superiority of UMAP for

preserving non-linear structures and achieving clearer topological separation. UMAP enables more informative visualization of transitional regions (e.g., noisy or hybrid datasets), which are common in real scRNA-seq data. Nonetheless, UMAP’s performance is sensitive to parameter choices (e.g., number of neighbors, minimum distance), whose optimal values are not always straightforward to determine.

From a practical standpoint, the use of a Random Forest classifier enhanced the precision of geometric classification compared to the k-nearest neighbors method. This highlights the potential of combining geometric descriptors with machine learning models to produce more robust decision-support tools for scRNA-seq analysis.

That said, the complete reproduction of the pipeline was not without challenges. We encountered multiple technical obstacles throughout the reimplementation. First, the original codebase was poorly documented, with several key functions (notably in `utils.py`) lacking detailed comments, requiring a substantial effort to reverse-engineer the logic. Second, the handling of large `.npy` files and the memory demands of geodesic distance calculations (via Diffusion Pseudotime) occasionally slowed down experiments or caused crashes. For example, the storage of the resulting computation in `.npy` files took more than 2 days, which slowed down our experiments, as we wanted to compare several different preprocessing phases with each others. Furthermore, several library dependencies were outdated or incompatible with our Python environment, necessitating manual patching — particularly for `Ripser` and `Scanpy`. Lastly, the real-world datasets varied in format and preprocessing history, which forced us to manually adapt several stages of the pipeline. One additional challenge was that we were not able to fully understand all the parameter configurations of the framework, which limited our ability to systematically explore their influence on the final geometric embedding. These issues, while eventually resolved or worked around, underscore the broader challenge of reproducibility in computational biology.

In summary, this study confirms the methodological soundness of the framework proposed by Lim and Qiu for characterizing scRNA-seq dataset geometry, while also identifying multiple avenues for refinement. A promising direction for future work would be to incorporate supervised biological information (e.g., marker gene expression, functional annotations) or explainability tools to better interpret model decisions. Furthermore, extending the pipeline to handle spatial or multimodal single-cell datasets could significantly broaden its relevance to modern biomedical research.

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